Four-colour Rado numbers of $a(x-y)=bz$: seven values with checked proofs

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Abstract

Let $\mathrm{R}_k(a,b)$ be the least n such that every k-colouring of $\{1,\dots,n\}$ has a monochromatic solution of $a(x-y)=bz$ ($\gcd(a,b)=1$). We determine seven four-colour values: $\mathrm{R}_4=461$, 741 and 1501 for $3(x-y)=4z$, $5(x-y)=4z$ and $6(x-y)=5z$, $\mathrm{R}_4=313$ for $4(x-y)=3z$, $\mathrm{R}_4=625$ for $5(x-y)=2z$ and for $5(x-y)=3z$, and $\mathrm{R}_4=2001$ for $x-y=5z$, none of which we found in the literature. The lower bounds are explicit colourings, each from a construction that works for every number of colours; the upper bounds are SAT computations on a small subset of $[n]$ split by cube-and-conquer, with every proof checked by the formally verified checker `cake_lpr`. We prove $\mathrm{R}_k(a(x-y)=(a-1)z)\ge a^k+a^{k-1}-2a+1$ for all $k,a\ge 2$; equality holds for $k=4$ and $a=3,4,5,6$, but fails for $k=5$ and $a=3,4$ ($\mathrm{R}_5(4(x-y)=3z)\ge 1401>1273$). We also collect lower bounds and evidence on a conjectured recursion $\mathrm{R}_4=b(\mathrm{R}_3+2a)+1$ for $a<b$.

1 Introduction

For positive integers a,b,k the *Rado number* $\mathrm{R}_k(a,b)$ is the least n such that every k-colouring of $[n]=\{1,\dots,n\}$ contains a monochromatic solution of $a(x-y)=bz$ (the numbers x,y,z need not be distinct). We always assume $\gcd(a,b)=1$; then every solution has the form

$$z=as,\qquad x-y=bs\qquad (s\ge 1).$$

Known four-colour values include $\mathrm{R}_4(1,1)=45$ (one more than the Schur number $S(4)=44$), $\mathrm{R}_4(2,3)=226$ [4], and the values listed in [2]. For three colours see [2, 1, 5].

Theorem 1.1.

$$\mathrm{R}_4\big(3(x-y)=4z\big)=461,\qquad \mathrm{R}_4\big(5(x-y)=4z\big)=741,\qquad \mathrm{R}_4\big(6(x-y)=5z\big)=1501.$$

Theorem 1.2.

$$\mathrm{R}_4\big(4(x-y)=3z\big)=313,\qquad \mathrm{R}_4\big(5(x-y)=2z\big)=625,$$
$$\mathrm{R}_4\big(5(x-y)=3z\big)=625,\qquad \mathrm{R}_4\big(x-y=5z\big)=2001.$$

For each value, the lower bound is an explicit colouring of $[n-1]$ (Section 3) and the upper bound is a computer proof that every colouring of $[n]$ fails (Section 2). None of the seven values is in Table 10 of [2], and we did not find them elsewhere. The values of Theorem 1.2 were first found by us with a SAT solver alone; version 2 of this note adds the checked proofs. They agree with the patterns $\mathrm{R}_4=a^4$ for $a\ge b+2$ (known for $(3,1)$, $(4,1)$, $(5,1)$ [2]), with Conjecture H3 below, and with $\mathrm{R}_4(x-y=(m-2)z)=m^4-m^3-m^2-m-1$, which [2] found for $4\le m\le 6$; here $m=7$.

Changes in version 2. New: Theorem 1.2 with checked proofs, Proposition 3.3, the five-colour bounds in Section 3, the second table of Section 5, the title. Everything else is unchanged.

2 Upper bounds: method

Lemma 2.1 (small set)**.** *Let* $K \geq 1$ *and*

$$S = \{\, am \leq n :\ m \leq K \,\} \ \cup\ \{\, x \leq n :\ x \equiv n \pmod b \,\}.$$

If S *has no* k*-colouring without monochromatic solutions inside* S*, then* $\mathrm{R}_k \leq n$*. Moreover we may assume that the* i*-th smallest element of* S *uses one of the colours* $0, \ldots, i-1$ *("first fit").*

Proof. A good colouring of $[n]$ restricts to S. Renaming the colours in order of first appearance gives a first-fit colouring. □

We take $K = \mathrm{R}_3$. The formula for S is about 25 times smaller than for $[n]$, but still too hard for a single solver run. We therefore use cube-and-conquer [3]:

1. The *root cubes* are all first-fit colourings of the first 7 elements of S.
2. A cube that is not solved within a time limit is split further by the next 3 elements of S.
3. Every leaf cube is shown unsatisfiable by CaDiCaL, which writes an LRAT proof for the formula with the cube added as unit clauses.
4. Every proof is checked by `lrat-check` and by the formally verified checker `cake_lpr` [6].
5. A separate program checks that the tree is complete, i.e. that the leaves cover all first-fit colourings.

In addition, every clause of the encoding is re-checked against the equation by an independent routine.

equation	n	$K = \mathrm{R}_3$	leaf cubes	checked by `cake_lpr`
$3(x-y) = 4z$	461	109	22863	all
$5(x-y) = 4z$	741	141	813	all
$6(x-y) = 5z$	1501	241	6833	all
$4(x-y) = 3z$	313	73	715	all
$5(x-y) = 2z$	625	125	715	all
$5(x-y) = 3z$	625	125	715	all
$x-y = 5z$	2001	286	3438	all

Here $\mathrm{R}_3(3,4) = 109 = b(b^2+b+2a+1)+1$ [1, 5]; $\mathrm{R}_3(5,4) = 141$, $\mathrm{R}_3(6,5) = 241$, $\mathrm{R}_3(4,3) = 73$ are the values $a^3+(a-1)^2$, $\mathrm{R}_3(5,2) = \mathrm{R}_3(5,3) = 125$ the values a^3 $(a \geq b+2)$, and $\mathrm{R}_3(x-y = 5z) = 286$ the value $m^3 - m^2 - m - 1$ with $m = 7$, all from [2].

3 Lower bounds

The value 461**.** The recursive colouring of [5, Lemma 4.1 and Theorem 4.5] for $(a, b) = (3, 4)$ and $k = 4$ colours [460] without monochromatic solutions. We also checked this colouring against all solutions.

The values 313**,** 741 **and** 1501**.** They are the case $k = 4$, $a = 4, 5, 6$ of the following theorem.

Theorem 3.1. *For all $k \geq 2$ and $a \geq 2$:* $\mathrm{R}_k(a(x-y) = (a-1)z) \geq a^k + a^{k-1} - 2a + 1$.

Proof. Let $n = a^k + a^{k-1} - 2a$, and let $v(x)$ be the exponent of a in x. Since $n < a^{k+1}$ we have $v(x) \leq k$.

The colouring. A number with $v(x) = j \geq 1$ gets colour c_j. For the numbers with $v(x) = 0$, cut $[1, n]$ into consecutive intervals

$$I_2,\ I_3,\ \ldots,\ I_{k-1},\ I_k,\ I'_{k-1},\ \ldots,\ I'_3,\ I'_2, \qquad |I_j| = |I'_j| = (a-1)a^{j-1},$$

and give the numbers with $v(x) = 0$ in $I_j \cup I'_j$ the colour c_j. The total length is $2(a^{k-1} - a) + (a-1)a^{k-1} = n$, and explicitly

$$I_j = [\,a^{j-1} - a + 1,\ a^j - a\,], \qquad I'_j = [\,n - a^j + a + 1,\ n - a^{j-1} + a\,].$$

Note that colour c_1 is used only for numbers with $v(x) = 1$.

No monochromatic solution. Let (x, y, z) be a solution of colour c_j, so $z = as$ and $x - y = (a-1)s$. Since $\gcd(a, a-1) = 1$,

$$v(z) = 1 + v(s), \qquad v(x-y) = v(s).$$

As $v(z) \geq 1$, the colour of z is $c_{v(z)}$, so $v(z) = j$ and $v(x-y) = j - 1$. Each of x, y has $v = j$, or $v = 0$ and $j \geq 2$.

- If $v(x) = v(y) = j$, then $v(x-y) \geq j$, a contradiction.
- If exactly one of x, y has $v = 0$, then $v(x-y) = 0$, so $j = 1$; but colour c_1 is not used for $v = 0$.
- So $v(x) = v(y) = 0$, $j \geq 2$, and $x - y = (a-1)a^{j-1}u$ with $a \nmid u$ and $z = a^j u \leq n$. If x and y lie in the same interval of colour c_j, then $x - y < (a-1)a^{j-1}$, a contradiction. Otherwise $j \leq k-1$, $y \in I_j$, $x \in I'_j$, and

$$x - y \ \geq\ (n - a^j + a + 1) - (a^j - a) \ =\ a^k + a^{k-1} - 2a^j + 1.$$

  - If $j \leq k-2$: $x - y = \frac{a-1}{a}\, z \leq \frac{a-1}{a}\, n < a^k - a^{k-2} \leq a^k + a^{k-1} - 2a^j + 1$, a contradiction.
  - If $j = k-1$: from $a^{k-1}u = z \leq n < a^{k-1}(a+1)$ we get $u \leq a$, and $a \nmid u$ gives $u \leq a-1$. Then $x - y \leq (a-1)^2 a^{k-2} < a^k - a^{k-1} + 1$, a contradiction. □

For $k = 3$ this construction is [2, Lemma 4.2], where equality is proved. We also checked the colouring against all solutions for $2 \leq a \leq 7$, $2 \leq k \leq 5$ (when $n \leq 20000$).

Conjecture 3.2 (H3)**.** $\mathrm{R}_4(a(x-y) = (a-1)z) = a^4 + a^3 - 2a + 1$.

This holds for $a = 3$ (103, [2]) and $a = 4, 5, 6$ (Theorems 1.1 and 1.2). For $a = 7$ the predicted value is 2731. The analogous statement for five colours is false: explicit colourings (found by SAT and checked against all solutions) give

$$\mathrm{R}_5(3(x-y) = 2z) \geq 399 > 319, \qquad \mathrm{R}_5(4(x-y) = 3z) \geq 1401 > 1273,$$

where 319 and 1273 are the values of Theorem 3.1 for $k = 5$. In the colouring of [1400] the numbers with $v(x) = 1$ form one colour class, but the other colours are not determined by $v(x)$.

The values 625**.** For coprime a, b the colouring of $[a^k - 1]$ by the exponent of a is good, so $\mathrm{R}_k(a, b) \geq a^k$ [2, Lemma 4.1]; for $a = 5$, $k = 4$ this gives 625. For $a \geq b + 2$, [2] proved $\mathrm{R}_3 = a^3$, and their Table 10 has $\mathrm{R}_4 = a^4$ for $(3, 1)$, $(4, 1)$, $(5, 1)$; Theorem 1.2 adds $(5, 2)$ and $(5, 3)$. Again five colours break the pattern: $\mathrm{R}_5(3(x - y) = z) > 320 > 3^5$ [5]. For $4(x - y) = z$ we searched for a 5-colouring of [1024] and did not find one (this is not an upper bound).

The value 2001**.** It comes from the following construction, which we include for completeness; it is a simple product construction and is probably known.

Proposition 3.3. *Let $b \geq 1$, $N_1 = b$ and $N_k = (b + 2)N_{k-1} + b$. Then $\mathrm{R}_k(x - y = bz) \geq N_k + 1 = \dfrac{b(b+2)^k + 1}{b+1}$ for every $k \geq 1$.*

Proof. For $k = 1$, colour $[1, b]$ with one colour: a solution needs $x - y = bz \geq b$, so $x \geq b+1$. Let C be a good $(k-1)$-colouring of $[1, N]$, $N = N_{k-1}$, put $s = (b+1)N + b$ and $n = s + N = N_k$, and colour $[1, n]$ as follows: $x \leq N$ gets $C(x)$; the middle block $B = [N + 1, s]$ gets a new colour; $x > s$ gets $C(x - s)$. Let (x, y, z) be a monochromatic solution, $x - y = bz$, $x > y$.

- New colour: $x, y, z \in B$, so $z \geq N + 1$ and $x - y \geq b(N + 1) = |B|$, impossible.
- $z > s$: then $bz \leq x - y \leq n - 1 = s + N - 1$ forces $b = 1$ (for $b \geq 2$, $bz \geq 2s + 2 > s + N$). Then $y = x - z \leq N - 1$ is in the bottom copy and $x > z > s$ in the top copy, and $(x - s) - y = z - s$ is a monochromatic solution for C.
- $z \leq N$: if x, y are in the same copy, shifting them gives a monochromatic solution for C. If $y \leq N$ and $x > s$, then $x - y \geq s + 1 - N = bN + b + 1 > bz$. □

For $b = 5$, $k = 4$ this is 2001. With $m = b+2$ the bound reads $((m-2)m^k+1)/(m-1)$; for $k = 3$ it is the value $m^3 - m^2 - m - 1$ proved in [2], and for $k = 4$ it is $m^4 - m^3 - m^2 - m - 1$. For $b = 1$ (Schur numbers) it is the classical bound $(3^k + 1)/2$, which is not sharp for $k = 4$ ($\mathrm{R}_4 = 45$). For $x - y = 2z$ we searched for a 5-colouring of [683] (one more than $N_5 = 682$) and did not find one (this is not an upper bound).

4 The case $a < b$: a recursion and its limits

In [5] a recursive colouring gives $R_k \geq L_k$ with $L_2 = b^2 + b + 1$, $L_k = b(L_{k-1} + 2a) + 1$ for every k, when $D = a^2 + ab - b^2$ satisfies $D \leq 1$ or $a + 1 \leq D \leq 2a$ (and $a^2 + (b-1)a \leq b^2 + b + 1$). For three colours L_3 is the conjectured value of R_3.

Conjecture 4.1 (K4)**.** *In the cases above,* $R_4(a, b) = b(R_3(a, b) + 2a) + 1$.

Evidence and limits:

- Equality holds for $(3, 4)$ (Theorem 1.1). The recursive colouring gives the further lower bounds

$R_4(3,5)$	$R_4(4,7)$	$R_4(5,8)$	$R_5(3,4)$	$R_5(3,5)$
≥ 961	≥ 3249	≥ 5401	≥ 1869	≥ 4836

  all of which are the values L_4, L_5 predicted by the recursion.

- If the multiples of a are fixed to the recursive pattern, the remaining constraints on the residue class $x \equiv n \pmod b$ become unsatisfiable exactly at $n = L_4$ for $(a, b) = (3, 4)$, $(3, 5)$, $(4, 7)$, $(5, 8)$, $(3, 7)$ and $(4, 9)$.

- For $(3, 5)$ Lemma 2.1 does not decide 961: S is colourable even with all multiples of 3 included, because in S the fourth colour can enter the copy $3 \cdot [1, 186]$ (from $m = 170$ in the colouring found). That colouring does not extend to [961], and all our structured searches for a colouring of [961] fail. A further cube-and-conquer run on a reduced set of 479 elements did not finish either: after splitting, every cube was either refuted within a fraction of a second or not decided within 300 to 1800 seconds. We expect $R_4(3, 5) = 961$ but have no proof; the upper bound is open.

- Outside the region the recursion fails: a colouring of [960] for $(4, 5)$ (found by SAT with the colour depending only on the a-adic valuation and $\lfloor x/2 \rfloor$, checked against all solutions) gives $R_4(4, 5) \geq 961 > 941$. In this colouring the fourth colour appears in the copy from $m = 156$ on.

- For $a = 2$ the values deviate already for three colours ($R_3(2, 3) = 61$, $R_3(2, 5) = 181$), and $R_4(2, 3) = 226 \neq 196$; for $(2, 5)$ an explicit colouring of [950] (found by SAT, checked against all solutions) gives $R_4(2, 5) \geq 951$, well above the value 926 predicted by the recursion.

Five colours: the method of Lemma 2.1 did not finish for $R_5(3, 4) \leq 1869$ (thousands of hard cubes).

5 Computing environment

All computations of this note ran on one desktop PC, except the search for the colouring of [950] for $(2, 5)$ and some other colourings of Section 4, which ran in a Linux cloud sandbox. The five-colour searches of Section 3 used CaDiCaL 3.0.1 on cubes of the first 6 free numbers, at most 300 seconds per cube.

CPU	Intel Core i5-11400F (6 cores, 12 threads)
Memory	32 GB DDR4
GPU	NVIDIA GeForce RTX 3060 (not used; the SAT solvers run on the CPU)
System	Windows 11 Home; solvers and proof checkers in WSL (Linux)
Workers	11 processes for the cube search and for the proof check
Cube search	MiniSat 2.2 through PySAT; cubes it could not solve quickly were tried by CaDiCaL 1.5.3 (PySAT) before splitting
Proofs	CaDiCaL 3.0.1 (static build, commit `c607304`), LRAT
Proof checkers	`lrat-check` (drat-trim, commit `2e3b2dc`) and `cake_lpr` (commit `a36874a`)

	461	741	1501
clauses per leaf formula	31,151–31,160	72,288–72,291	191,958–191,964
leaf cubes, all UNSAT	22,863	813	6,833
proofs accepted by `cake_lpr`	22,863	813	6,833
total size of the LRAT proofs	196 GB	24.5 GB	256 GB
cube search, first to last record	12 h 56 min	34 min	3 h 9 min
proof check, task time (sum over workers)	34 h	2 h	36 h

	313	625, $b = 2$	625, $b = 3$	2001
clauses per leaf formula	22,266	130,288	79,005	325,818–325,824
leaf cubes, all UNSAT	715	715	715	3,438
proofs accepted by `cake_lpr`	715	715	715	3,438
total size of the LRAT proofs	1.6 GB	7.1 GB	15.4 GB	106 GB
cube search, first to last record	28 min	28 min	32 min	3 h 48 min
proof check, task time (sum over workers)	0.3 h	1.0 h	1.6 h	19 h

The four searches of the second table ran together, on 3 November 2026, without interruption. The runs for the first table were stopped and resumed several times, so the times in its last two rows are approximate: "first to last record" includes pauses, and "task time" is the sum of the times of the single tasks over all workers. The proofs were written to a temporary folder and deleted after checking; they can be regenerated with the published scripts.

6 Reproducibility

Encodings, cube trees, results of the proof checks, checkers and colourings are available at `https://github.com/IchiTaku326/rado-ab` (release v2). The LRAT proofs themselves are not stored (they are large); they can be regenerated with the published scripts and checked again. The checker `cake_lpr` was compiled with heaps of 4 and 14 GB.

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